

Unit Test on RF and Microwave Engineering
Unit Test 1 (Chapter 2)

Pass Mark: 10
Fail: 4

Time: 50 mins

Set 1

Question 1 (10 Marks): A load impedance $100-j80$ ohm is connected to a transmission line. Find the size and placement of the matching stub that will remove all the standing waves and match load to the line. Use single stub series tuning short circuited stub. Draw its electrical diagram and physical connection.

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Set 2

Question 1 (10 Marks): A 75 ohm lossless transmission line operating at $f=1.8\text{GHz}$ is terminated with $Z_L=40+j60$ ohm. Find the size and placement of the matching stub that will remove all standing waves and match the load to the line. Use single stub shunt tuning. Draw its electrical diagram and physical connection.

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Set 3

Question 1 (10 Marks): 50 ohm lossless transmission line operating at $f=2.5\text{GHz}$ is terminated with $Z_L=90-j150$ ohm. Find the size and placement of the matching stub that will remove all standing waves and match the load to the line. Use single stub series tuning. Draw its electrical diagram and physical connection..

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Set 4

Question 1 (10 Marks): 50 ohm lossless transmission line operating at $f=3\text{GHz}$ is terminated with $Z_L=25+j90$ ohm. Find the size and placement of the matching stub that will remove all standing waves and match the load to the line. Use a single stub shunt tuning stub. Draw its electrical diagram and physical connection.